Sangjun (Samuel) Ko

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Education

University of Illinois Urbana-Champaign — Champaign, Illinois
B.S. Statistics, Mathematics Minor; GPA: 3.93/4.00; Highest Distinction — January 2023 – May 2025

University of Michigan–Shanghai Jiao Tong University Joint Institute — Shanghai, China
Mechanical Engineering coursework; transferred to UIUC — September 2019 – December 2020

Experience

Kimberly-Clark — North America Family Care R&D — Neenah, Wisconsin
Statistician (contract assignment through Staffmark Group) — June 2025 – Present

- Validate measurement methods before decisions rely on them: Gage R&R with variance components, inter-rater agreement (Fleiss' and Cohen's kappa), and blinded attribute validations scored like sensitivity and specificity.
- Design and analyze factorial and repeated-measures experiments (mixed models, nonparametric tests, Holm-adjusted contrasts), including screening designs that relate benchtop tests to consumer home-use data and a confirmation study in which no option won everywhere; implausible runs were flagged and documented rather than deleted.
- Designed balanced equivalence studies of candidate replacement hardware across five products; found bias that grew with product slough level (up to 9 mg against a ± 0.7 mg working reference) and drift in the reference hardware, so the data did not support a drop-in replacement, and shared the evidence with the supplier.
- Decomposed variance across 624 destructive tab-tear pulls and set a statistical guardband for a size that failed precision criteria instead of loosening the specification.
- Rebuilt the product-performance modeling engine from legacy SAS into a Python/Flask web app, resolving split-plot error structures; validated against ten historical studies (nine matched; one difference documented).
- Analyzed multi-night actigraphy from 58 infants in a randomized incomplete-block home-use study (20+ sleep metrics) with mixed-effects models (infant random effects); logged data exclusions and reported "not detected," not "no difference," to the product team and an external clinical sleep specialist, stopping an unsupported claim.

University of Illinois Urbana-Champaign — Champaign, Illinois
Undergraduate Course Assistant — Data Science Discovery and Professional Statistics — August 2023 – May 2025

- Mentored 100+ students in data analysis and experimental design; in Professional Statistics, a statistical consulting course, graded consulting assignments and gave feedback to 20+ students.
- Managed Git-based course content for a platform serving 1,000+ students.

Applied Technologies for Learning in Arts & Sciences (ATLAS), UIUC — Champaign, Illinois
AI/ML Intern Team Leader — January 2024 – December 2024

- Led five interns building predictive models on datasets with 100,000+ records; on a graduation-prediction project (50,000+ student records), feature engineering and outlier review raised AUC from 0.65 to 0.82.

Korean Augmentation to the U.S. Army (KATUSA) — Dongducheon, South Korea
Soldier (mandatory military service) — June 2021 – December 2022

- Served 18 months with U.S. Army units (tank crew, then supply and inventory); translated for 2,000+ U.S. and Korean soldiers during joint training exercises.

Honors and Service

3rd Place, COVID Information Commons Student Paper Challenge — September 2024
Co-authored paper on the socioeconomic and mental-health effects of COVID-19 in Mexico (ENCOVID-19 survey, 1,400+ respondents; k-nearest-neighbors imputation, geospatial analysis)

Army Achievement Medal, U.S. Army — December 2022

Volunteer math tutor, Appleton Public Library — September 2025 – Present

Technical Skills

Statistics: Design of experiments, mixed-effects and split-plot models, variance components, Gage R&R, kappa, Bland-Altman, multiplicity correction

Software: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), R, SAS, JMP, SQL, Flask, Git

Languages: Korean and Spanish (native); English (fluent)